UNIVERSITI MALAYSIA PERLIS

INNOVATION IN EDUCATION · SHARING SESSION

Transforming Course Administration

An Automated Management Ecosystem for the NMK31704 Design Project

PRESENTER

Ts. Dr. Mohd Zamri Zahir Ahmad

TEAM MEMBERS

Ts. Dr. Ahmad Ashraf Abdul Halim · Ts. Dr. Aini Syuhada Md Zain · Dr. Nurol Husna Che Rose

Faculty of Electronic Engineering & Technology (FKTEN), UniMAP

DATE

14 October 2026

EVENT

Siri Perkongsian Kepimpinan Instruksional

Bulan Akademia 2026

ORGANISED BY Program Kerjasama Pusat Pembangunan Kepimpinan Pendidikan Lestari (LEAD) dan Persatuan Kakitangan Akademik (PKA)

THE COURSE

NMK31704 Design Project

- **Group-based design project** offered every Semester 2
- **100% continuous assessment**, no final examination
- **4 to 5 students** in a group, mixed across programmes
- **1 or 2 supervisors** for each group, plus **2 panels** for the final assessment
- **14 weeks**, run by **4 course coordinators**

THREE ENGINEERING TECHNOLOGY PROGRAMMES, ONE COURSE

UR6523007	Bachelor of Electronic Engineering Technology (Electronic Systems) with Honours
UR6523008	Bachelor of Electronic Engineering Technology (Electronic Telecommunication Design) with Honours
UR6523009	Bachelor of Electronic Engineering Technology (Electronic Network Design) with Honours

Students, supervisors and panels all come from these three programmes, so every decision has to be balanced across them.

THE SCALE

One semester, in numbers

182
students

43
project groups

80+
lecturers as supervisors and panels

592
weekly logbooks to verify and mark

258
evaluations to collect from supervisors and panels

12
rubrics feeding 7 course outcomes

Semester 2, 2025/2026. Evaluations counted as six per group: report by the supervisor and two panels, presentation by two panels, supervisor observation.

THE WORK

What a coordinator has to get right

BEFORE THE SEMESTER

Set up

- Form groups balanced by CGPA, programme, gender and race
- Pair supervisors from different programmes
- Keep lecturers with management posts as co-supervisor only
- Appoint panels who do not supervise the group

EVERY WEEK, FOR 14 WEEKS

Run

- Each group plans a Gantt chart
- Each group submits a weekly logbook
- The supervisor verifies it
- The coordinator checks that progress follows the plan

END OF THE SEMESTER

Assess

- Coordinators mark every logbook
- Students review their own group members
- Supervisors and panels mark the report and presentation
- Marks become grades and CO/PO attainment

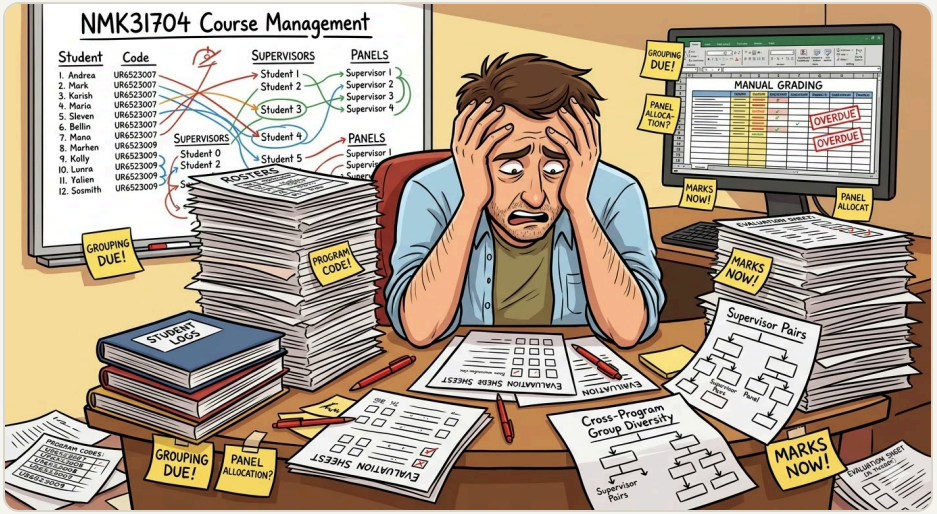
PART 1

The problem

Two semesters of running all of this by hand

THE PROBLEM

For two semesters, all of it was manual



- ✕ **Grouping:** student records sorted by hand, then checked again for every rule
- ✕ **Logbook:** written in MS Word, saved as PDF, signed by the supervisor, submitted in eLearning
- ✕ **Peer review:** Google Form, then manual queries in Google Sheets to filter and sort each student's scores
- ✕ **Assessment:** a Google Sheet made to look like a system. One sheet per group, so supervisor and panels could see each other's marks, and every panel change was edited by hand
- ✕ **Final materials:** each group sent its report, poster and video to its panels by email and eLearning

Semester 2, 2023/2024
Manual

Semester 2, 2024/2025
Manual

Semester 2, 2025/2026
Automated ecosystem

THE PROBLEM

Everyone carried part of the burden

Coordinators

Download files one by one, open each, record each mark. Filter and sort peer review scores student by student. Compile everything under deadline.

Supervisors

Receive a document from every group, sign it and send it back. No single place to see how a group is progressing.

Panels

Search email for each group's report, poster and video. Mark in a shared sheet where other evaluators' marks are visible.

Students

Prepare the logbook, convert it, get it signed, upload it. Send final materials to several lecturers and hope nothing is missed.

> **The work was not difficult. It was repeated hundreds of times, and every repetition was a chance for an error.**

PART 2

The innovation

Automate

grouping, supervisor and panel allocation

Centralise

the whole continuous assessment workflow

Streamline

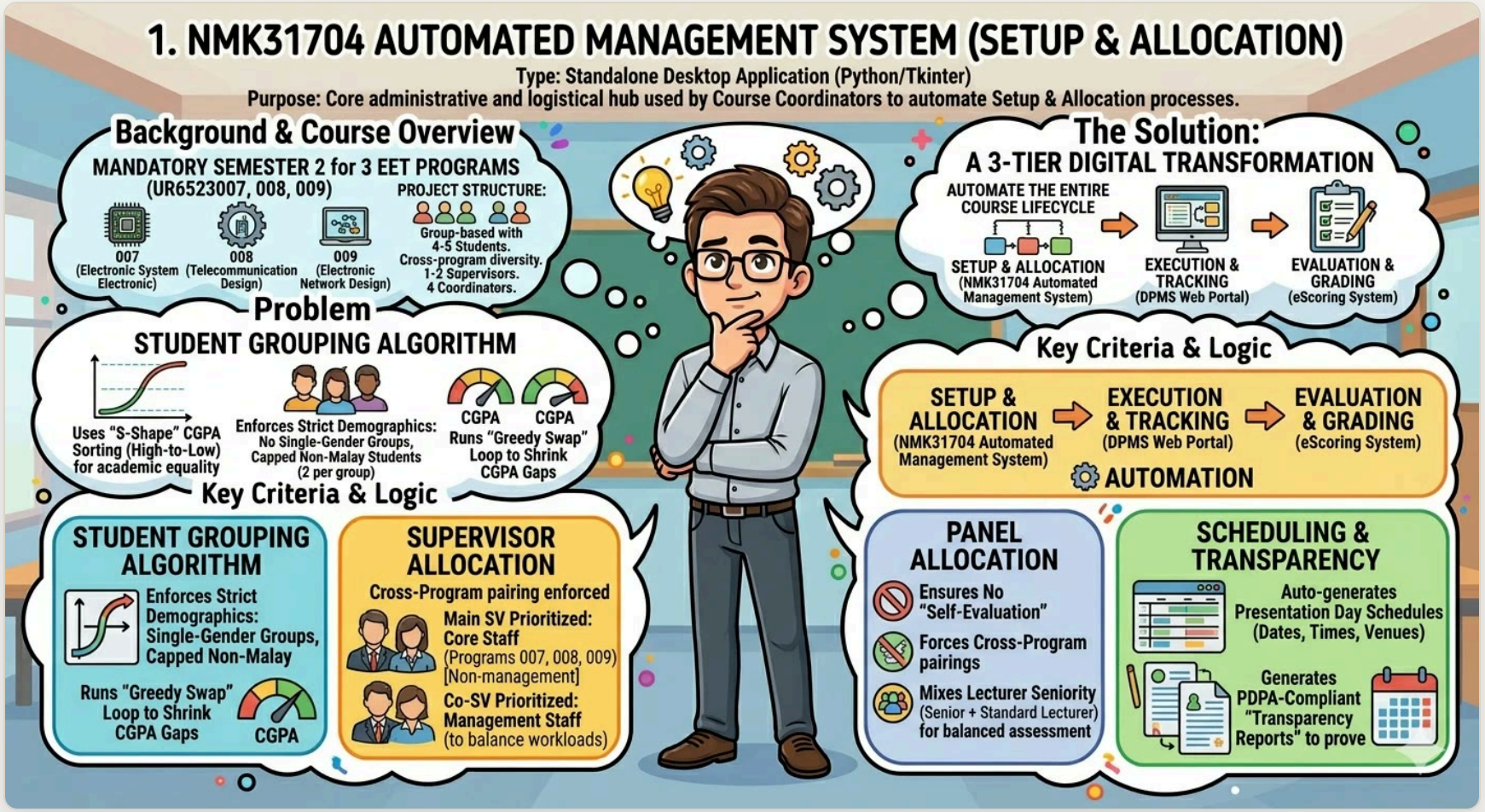
final evaluation with digital rubrics

One ecosystem, three systems, at a glance

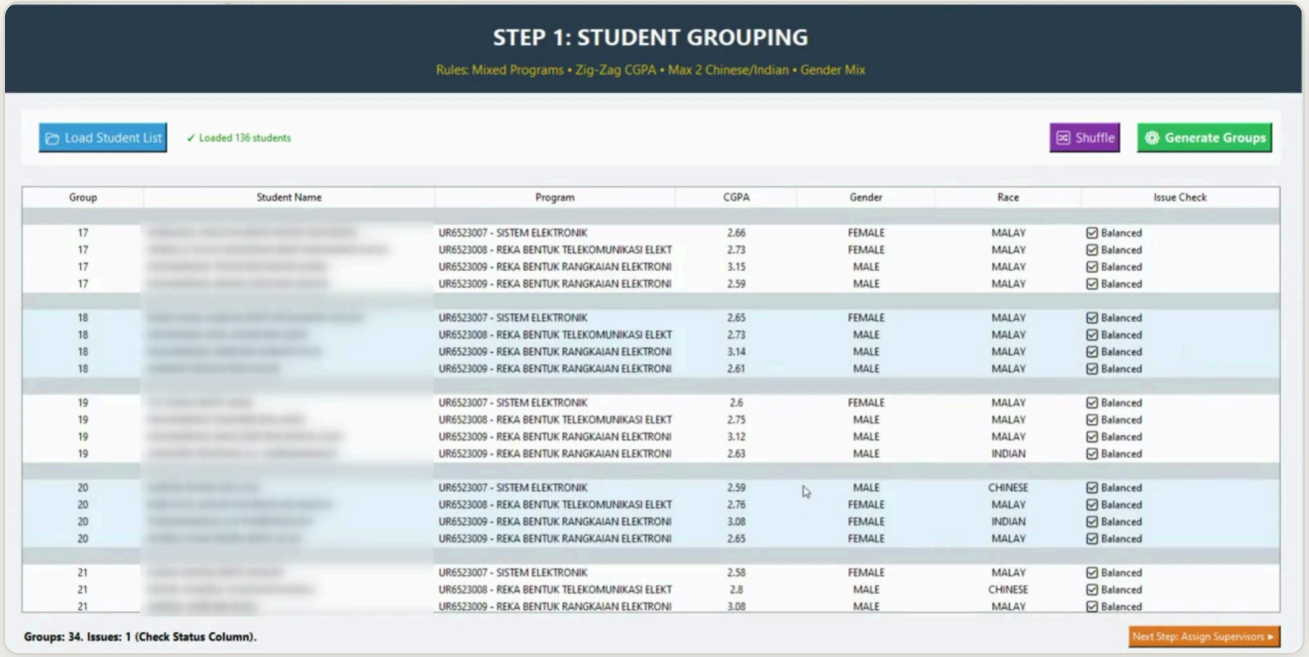
1 · SET UP
Automated Management System
Desktop application

2 · RUN
Design Project Management System (DPMS)
Web application

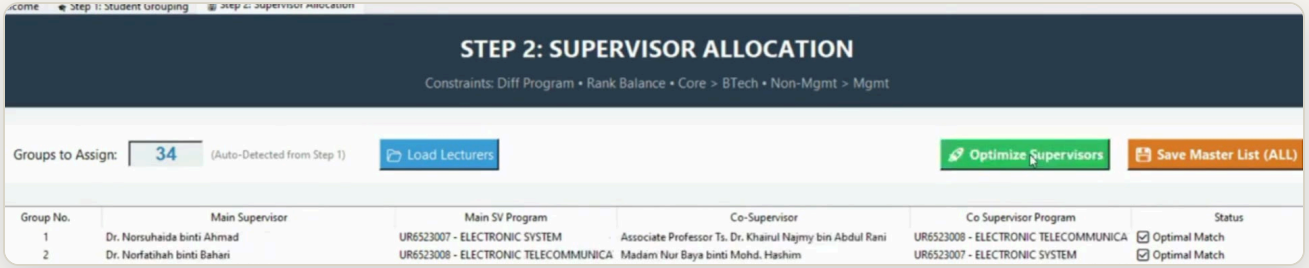
3 · ASSESS
eScoring System
Web application



Balanced groups and fair allocation in minutes



Step 1: load the student list, generate groups, every row checked. Click to enlarge.



Step 2: supervisor pairs from different programmes

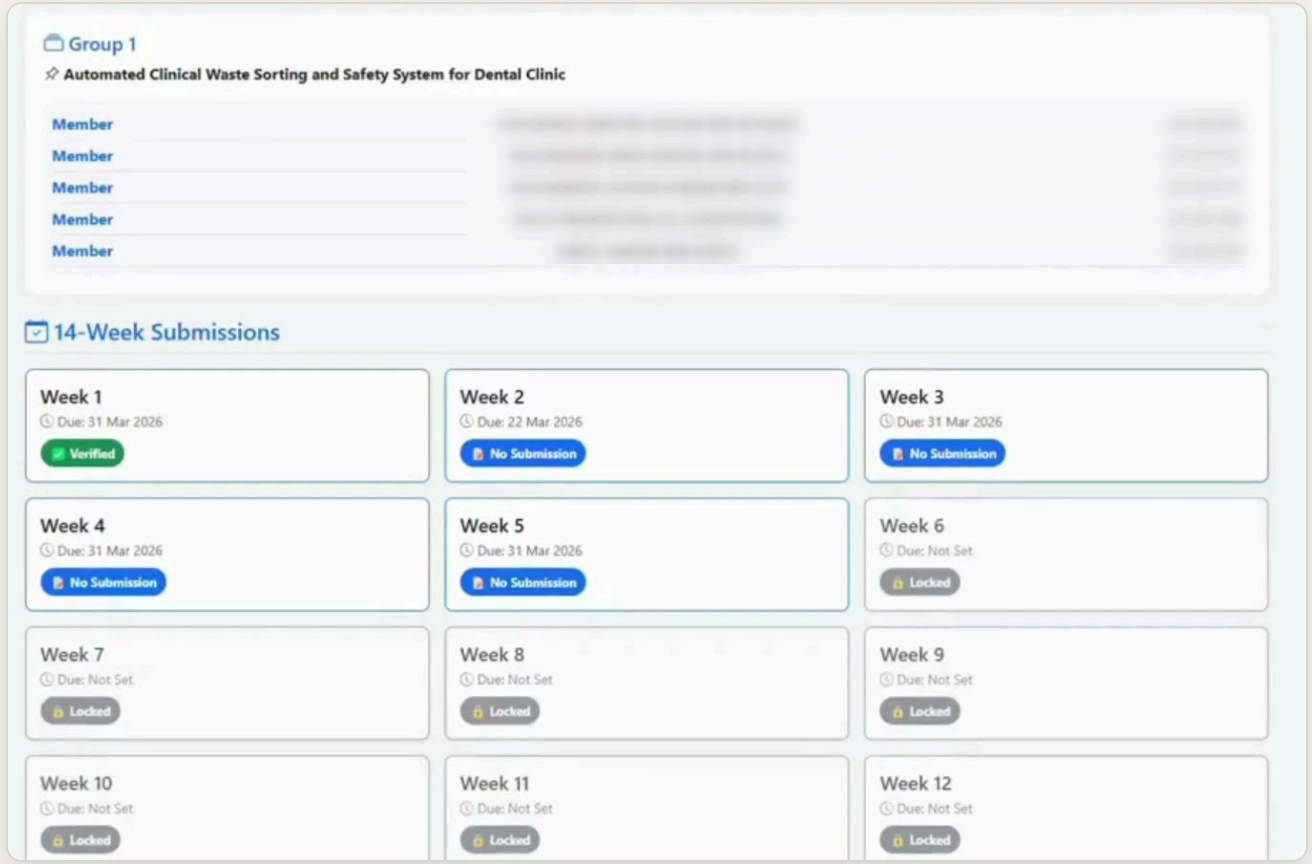
RULES THE SYSTEM ENFORCES

- ✓ Every group mixes the three programmes
- ✓ CGPA, gender and race balanced across groups
- ✓ Supervisor and co-supervisor from different programmes
- ✓ Management staff assigned as co-supervisor
- ✓ No panel evaluates a group they supervise

HOW IT WORKS

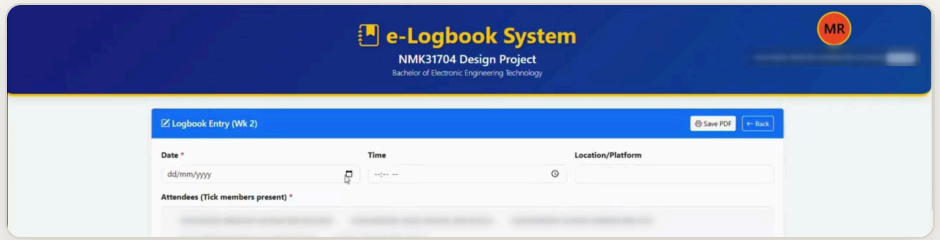
Students are ordered by CGPA and dealt into groups in a zig-zag. The system then swaps students between groups until the gaps are small, and reports how the allocation was made.

DPMS: one place for the whole 14 weeks



A student's page: the group, and the status of each week's logbook. Click to enlarge.

- ✓ **Weekly logbook** filled in on the web, with photos as evidence
- ✓ **Gantt chart** planned once, then ticked through the logbook each week
- ✓ **Supervisor verification** on screen, with email notification
- ✓ **Peer review** confidential, with marks calculated by the system
- ✓ **Final submission** of report, poster, brochure and video link
- ✓ **Coordinators mark logbooks** inside the same system



The weekly logbook form

eScoring: mark on the spot, track it live

EVALUATOR
My assigned groups
Role, venue, session, countdown

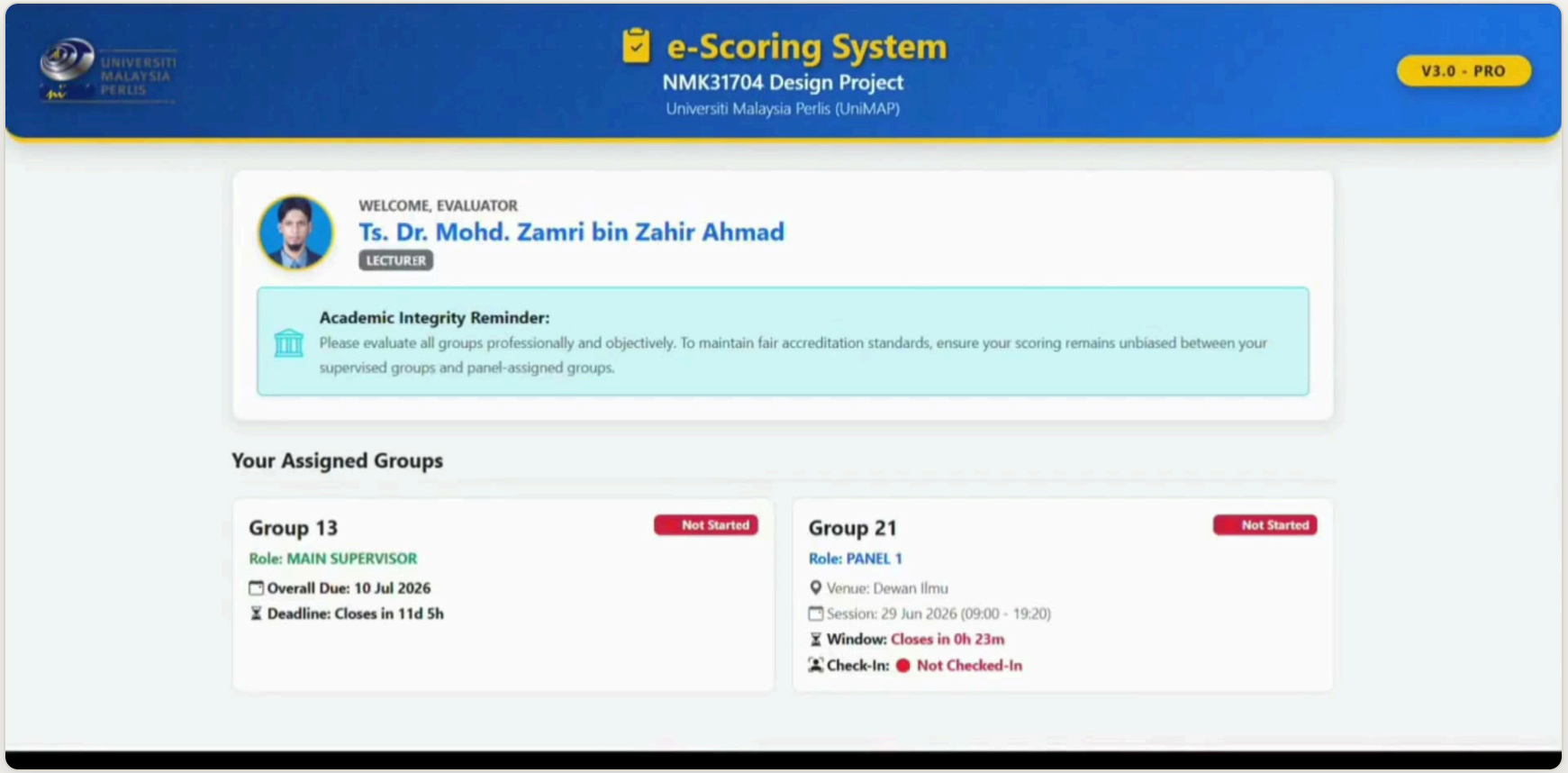
EVALUATOR
Check-in at the venue
Unlocks presentation marking

EVALUATOR
Marking with the rubric
Guide and live mark per criterion

EVALUATOR
Completed
Status and live marks summary

COORDINATOR
Evaluation status
Supervisor and both panels, per group

COORDINATOR
Live check-in monitor
Who has arrived, by venue



PART 3

The impact

What changed, and what the users said

THE IMPACT

Before and after

TASK	MANUAL, TWO SEMESTERS	WITH THE ECOSYSTEM
Form 43 balanced groups, assign supervisors and panels	Sorted by hand in a spreadsheet **about 2 to 3 working days**	Generated and checked by rules **minutes**
Collect, verify and mark 592 logbooks	Word, PDF, signature, eLearning, download one by one **about 30 hours of handling**	Submitted, verified and marked on one page **no file handling**
Peer review marks	Google Form, then filter and sort about 600 responses student by student	Rated in the system, marks calculated **instant**
Final materials to the panels	About 170 files sent by email and eLearning	Panels open them from the group page
Marks, grades, CO/PO and audit evidence	One shared sheet per group, queries rebuilt by hand **about 1 to 2 working days**	Private to each evaluator, calculated on submission **instant**
Who has not submitted or marked	Ask, wait, ask again	Live status for every group and evaluator

Counts are from Semester 2, 2025/2026. Time figures and the number of peer review responses are estimates by the coordinators, not measured values.

THE IMPACT

What 41 users told us

EASE OF USE FOR THEIR TASKS

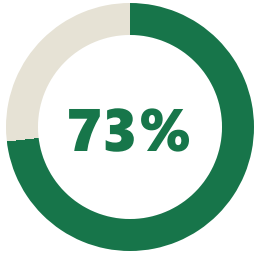
8.5 / 10

33 of 41 gave 8 or higher

STREAMLINES MANAGEMENT AND ASSESSMENT

8.6 / 10

35 of 41 gave 8 or higher



NO ISSUES WHILE USING IT

30 of 41. The other 11 reported an issue, mostly signing in and phone use.

MOST USEFUL FEATURE

Online logbook named by 18 of 31 who answered, then the **Gantt chart** and **supervisor verification**.

WHAT THEY ASKED FOR NEXT

Better phone support, more reliable sign-in, autosave for logbook drafts.

> *"Overall the system makes students work much easier."*

> *"Easy for students to update progress and supervisors can monitor project progress."*

> *"Good implementation to be used in future."*

Source: DPMS User Experience Survey on forms.app, 41 responses (34 students, 6 lecturers, 1 not stated), 29 May to 23 June 2026. share.forms.app/mohdzamrizahirahmad/design-project-management-system-dpms-dpms-user-experience-survey

FOR YOUR OWN COURSE

Built with what the university already gives us

WHAT IT RUNS ON

- **Google Apps Script** for the two web applications
- **Google Sheets** as the database
- **UniMAP Google account** for sign-in, so no passwords to manage
- **Python** for the grouping tool
- **No licence or hosting cost**

WHAT WE WOULD TELL ANOTHER COORDINATOR

1. **Start with the most repeated task.** For us it was the weekly logbook.
2. **Write your rules down first.** If a rule can be stated, it can be automated.
3. **Put the rubric online and show the marks live.** Evaluators see every criterion with its guide, and the mark as they score.
4. **Enter data once.** Every time the same list is retyped, an error appears.
5. **Run it for one semester, listen, improve.** The users will tell you what is missing.

RECOGNITION

Gold Award, EDIC 2026



Gold Award

Education Innovation Challenge (EDIC) 2026

International Conference on Quality, Sustainability & Innovation in Science and Technology Education (IQ-SITE 2026)

CATEGORY

Transformative Teaching

DATE

1 September 2026

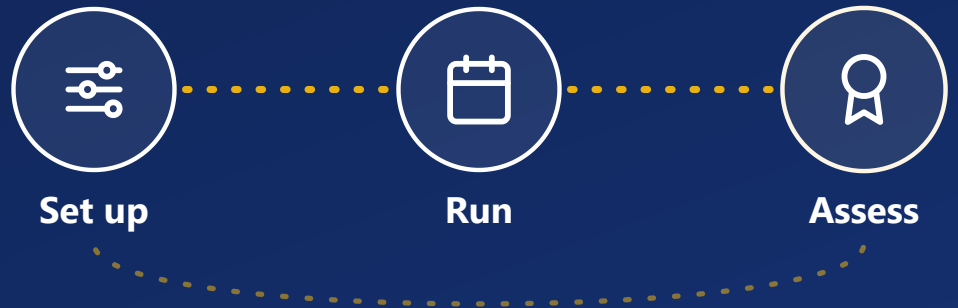
TEAM

Mohd Zamri Zahir Ahmad, Ahmad Ashraf Abdul Halim, Nurol Husna Che Rose, Aini Syuhada Md Zain

Thank you

Questions and discussion

These slides



The DPMS site

THE TEAM

Ts. Dr. Mohd Zamri Zahir Ahmad · Ts. Dr. Ahmad Ashraf Abdul Halim
Ts. Dr. Aini Syuhada Md Zain · Dr. Nurol Husna Che Rose

SEE IT

sites.google.com/unimap.edu.my/dpfkten
youtu.be/Z26XAPRhVqI (6-minute video)

Next: the three systems become one unified DPMS for Semester 2, 2026/2027.